ASHKBIZ DANEHKAR

Astrophysicist

Contact

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github.com/danehkar

Skills

- Computational Astrophysics
- Observational Astrophysics
- Theoretical Astrophysics
- Scientific Computing
- Data Analysis
- Data Visualization
- Data Science

Computing Skills

- C++
- Python
- Fortran
- IDL/GDL
- OpenMPI
- AstroPy
- SciPy
- Matplotlib
- NumPy
- pandas

References

Available Upon Request.

Typeset in HTML & CSS
See git.io/f4dXp

Summary

Astrophysicist with lots of experience in data analysis and visualization.
Research interest in AGN, starburst feedback, ISM astrophysics, and nebulae.

Experience

Scientist (Astronomy) — 2025 – Present
Universities Space Research Association

- Support Interagency Implementation and Advanced Concepts Team.

Research Scientist (Astrophysics) — 2022 – 2025
Eureka Scientific Inc

- Conducted research on the black hole spins in radio-quiet AGN.

Research Fellow — 2019 – 2021
University of Michigan

- Conducted hydrodynamic simulations of starburst feedback.
- Produced photoionization models of starburst-driven outflows.

Postdoctoral Fellow — 2015 – 2018
Harvard-Smithsonian Center for Astrophysics

- Analyzed X-ray data of active galactic nuclei and symbiotic systems.
- Conducted Hubble imaging analysis on ionized nebulae.

Visiting Early-Stage Researcher — 2008

- Investigated coupling between dual gravity and a BF model.

Education

Ph.D., Physics and Astronomy — 2014
Macquarie University, Australia

M.S., Plasma Physics (with distinction) — 2009
Queen's University Belfast, UK

M.S., Computational Electrical Engineering — 2007
University of Rostock, Germany

Grants

NASA NICER GO Program (80NSSC23K1098) — 2024
NASA Astrophysics Data Analysis Program (80NSSC22K0626) — 2022 – 2024

Awards

Macquarie University Research Excellence Scholarship — 2010 – 2014
Department for Employment and Learning Studentship — 2008 – 2009
Marie Curie Early Stage Researcher — 2008